

Education

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| • M.Sc. Data Science , Monash University, Australia. | 2024–2026 |
| • B.Sc. (Hons.) Computer Science , University of Nottingham, UK. | 2019–2020 |
| • B.Sc. Computer Science , Bina Nusantara International, Indonesia. | 2016–2020 |

Work Experience

Associate Data Scientist	tiket.com	Oct 2022–July 2024
Jakarta, Indonesia		

tiket.com is one of Indonesia's largest online travel agency companies, offering an app-based platform for booking flights, hotels, trains, car rentals, and entertainment tickets.

- Upgraded the hotel image quality prediction accuracy by 10% by integrating advanced OpenCV preprocessing and deep learning techniques, leading to more reliable data inputs for machine learning models on GCP infrastructure.
- Engineered a mathematical-based user payment recommendation predictive model using Numpy and Scikit-learn that boosted payment conversion rate by 4.8%, translating into an additional IDR 94 billion in gross booking value (GBV).
- Developed a 90% accuracy CatBoost-based model to tackle the hotel room grouping problem through extensive feature engineering and similarity scoring.
- Optimized the existing hotel recommendation engine utilizing POI and geospatial analytics using ensemble models, resulting in an IDR 8 billion increase in gross booking volume (GBV) and generating 1.1 billion IDR of margin.
- Streamlined flight reschedule identification process by developing an LSTM Tensorflow-based model, reducing 165 million IDR of operational expenses and 150 million IDR of RPA bot costs annually.

Projects & Achievements

Slot Filling & Intent Detection	Monash University	Sep 2025-Oct 2025
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- Integrated a fully automated Streamlit web app deployment that simultaneously predicts from 21 unique intents and 121 unique slot labels using shared representations.
- Performed multiple model comparisons from Baseline (CRF + Random Forest) to state-of-the-art models like JointBiLSTM, JointBiLSTM-Attention, and JointBERT with +95% intent accuracy and +80% slot F1-score.

Stock Performance Prediction	Monash University	Aug 2025-Sep 2025
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- Developed an Ensemble model to predict 2 tasks: whether a stock will achieve a positive or negative excess return (classification) and the continuous excess return (regression).
- Achieved 85% F1-scores and 0.035 RMSE on the test data.
- Placed 6th place in a private Kaggle competition out of 71 participants with a final RMSE of 0.103.

Melbourne Air Quality-Pedestrian Analysis	Monash University	May 2025-June 2025
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- Developed a visualization web app using HTML, CSS, and D3.js, which was fully deployed on Vercel.
- Presented the history of 5 different air pollutants: CO, NO₂, O₃, PM2.5, PM10 on different levels of granularity and pedestrian activity across 80+ different areas in Melbourne.
- Preprocessed 1 year of air quality records (~765,000 records) in 2022, and more than 8,700 records of pedestrian activity.

Pokémon Battle Prediction	Monash University	April 2025-May 2025
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- Built and deployed an interactive RShiny app for real-time battle prediction for a total of 800+ Pokémon from Gen 1-7 using extensive features such as type effectiveness, movesets, and Pokémon stats.
- Achieved an overall prediction accuracy of 90% using an XGBoost model on the test set.

Technologies and Languages

- Languages : Python, R, C++, SQL
- Technologies: Tools: BigQuery, MongoDB, Apache Spark, Kafka, Tableau, Git, Jupyter Notebook, GCP
- Libraries : Pandas, NumPy, Scikit-learn, XGBoost, PyTorch, TensorFlow, Matplotlib, Seaborn